

1. Instalación de Terraform

Se instaló Terraform en Windows usando el gestor de paquetes winget, que descarga e instala automáticamente el binario y lo agrega al PATH del sistema:

```
winget install HashiCorp.Terraform
```

```
Windows PowerShell
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Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

PS C:\Users\robin> winget install HashiCorp.Terraform
Encontrado HashiCorp.Terraform [Hashicorp.Terraform] Versión 1.14.7
El propietario de esta aplicación le concede una licencia.
Microsoft no es responsable, ni tampoco concede ninguna licencia de paquetes de terceros.
Descargando https://releases.hashicorp.com/terraform/1.14.7/terraform_1.14.7_windows_amd64.zip
29.3 MB / 29.3 MB
El hash del instalador se verificó correctamente
Extrayendo archivo...
Archivo extraído correctamente
Iniciando instalación de paquete...
Variable de entorno PATH modificada; reinicie el shell para usar el nuevo valor.
Alias de línea de comandos agregado: "terraform"
Instalado correctamente
PS C:\Users\robin> |
```

Verificación de la instalación

Se verificó que Terraform quedó correctamente instalado ejecutando el comando de versión:

```
terraform -version
```

```
Windows PowerShell
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Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

PS C:\Users\robin> terraform -version
Terraform v1.14.7
on windows_amd64
PS C:\Users\robin>
```

2. Instalación de Azure CLI

Se instaló el Azure CLI usando winget para poder autenticarse y gestionar recursos desde la línea de comandos:

```
winget install Microsoft.AzureCLI
```

```
PS C:\Users\robin> winget install Microsoft.AzureCLI
Se encontró un paquete existente ya instalado. Intentando actualizar el paquete instalado...
Encontrado Microsoft.AzureCLI [Microsoft.AzureCLI] Versión 2.84.0
El propietario de esta aplicación le concede una licencia.
Microsoft no es responsable, ni tampoco concede ninguna licencia de paquetes de terceros.
Descargando https://azcliprod.blob.core.windows.net/msi/azure-cli-2.84.0-x64.msi
64.8 MB / 64.8 MB
El hash del instalador se verificó correctamente
Iniciando instalación de paquete...
Instalado correctamente
Notas: Winget installs the 64-bit CLI on 64-bit OS by default now. If you have used the 32-bit CLI before, please follow this guide to migrate to 64-bit version: https://learn.microsoft.com/cli/azure/install-azure-cli-windows#migrate-to-64-bit-azure-cli
```

Verificación

Se verificó la instalación comprobando la versión instalada:

```
az version
```

```
Windows PowerShell
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Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

PS C:\Users\robin> az version
{
  "azure-cli": "2.84.0",
  "azure-cli-core": "2.84.0",
  "azure-cli-telemetry": "1.1.0",
  "extensions": {}
}

[Survey] Tell us what you think of Azure CLI. This survey should take about 2 minutes. Run 'az survey' to open in browser. Learn more
at https://go.microsoft.com/fwlink/?linkid=2203309

PS C:\Users\robin>
```

3. Autenticación en Azure

Se ejecutó az login para autenticarse con la cuenta universitaria y obtener acceso a la suscripción Azure for Students de la Escuela Colombiana de Ingeniería Julio Garavito:

```
az login --tenant "50640584-2a40-4216-a84b-9b3ee0f3f6cf"
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

Instale la versión más reciente de PowerShell https://aka.ms/PSWindows

PS C:\Users\robin> az version
{
  "azure-cli": "2.84.0",
  "azure-cli-core": "2.84.0",
  "azure-cli-telemetry": "1.1.0",
  "extensions": {}
}

[Survey] Tell us what you think of Azure CLI. This survey should take about 2 minutes. Run 'az survey' to open in browser. Learn more
at https://go.microsoft.com/fwlink/?linkid=2203309

PS C:\Users\robin> az login
Select the account you want to log in with.
=2271136
```

Iniciar sesión

Vamos a iniciar sesión

Usar una de estas cuentas

- robinson.kawaki677@outlook.es
Cuenta de Microsoft
- robinson.nunez-p@mail.escuelaing.edu.co
Agregar una cuenta profesional o educativa

Usar otra cuenta

- Cuenta de Microsoft
- Correo electrónico, teléfono o Skype
- Agregar una cuenta profesional o educativa
Asignada por su organización

Continuar

Se verificó que la suscripción quedó activa y vinculada correctamente al tenant de la ECI:

```
PS C:\Users\robin> az login
Select the account you want to log in with. For more information on login with Azure CLI, see https://go.microsoft.com/fwlink/?linkid=2271136

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No  Subscription name  Subscription ID  Tenant
---  -
[1] * Azure for Students  056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a  ESCUELA COLOMBIANA DE INGENIERIA JULIO GARAVITO

The default is marked with an *; the default tenant is 'ESCUELA COLOMBIANA DE INGENIERIA JULIO GARAVITO' and subscription is 'Azure for Students' (056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a).
```

Se guardó el Subscription ID ya que será necesario para configurar GitHub Actions en pasos posteriores:

```
Select a subscription and tenant (Type a number or Enter for no changes): 1

Tenant: ESCUELA COLOMBIANA DE INGENIERIA JULIO GARAVITO
Subscription: Azure for Students (056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

PS C:\Users\robin> az account show
{
  "environmentName": "AzureCloud",
  "homeTenantId": "50640584-2a40-4216-a84b-9b3ee0f3f6cf",
  "id": "056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a",
  "isDefault": true,
  "managedByTenants": [],
  "name": "Azure for Students",
  "state": "Enabled",
  "tenantDefaultDomain": "pruebacorreoescuolaingeducu.onmicrosoft.com",
  "tenantDisplayName": "ESCUELA COLOMBIANA DE INGENIERIA JULIO GARAVITO",
  "tenantId": "50640584-2a40-4216-a84b-9b3ee0f3f6cf",
  "user": {
    "name": "robinson.nunez-p@mail.escuelaing.edu.co",
    "type": "user"
  }
}
PS C:\Users\robin> |
```

Subscription ID

056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a

4. Creación del Backend Remoto

El backend remoto permite que el estado de Terraform - terraform.tfstate se almacene en Azure Storage en lugar de localmente. Esto habilita el state locking, evitando conflictos cuando múltiples personas trabajan sobre la misma infraestructura.

4a. Resource Group

Se creó el Resource Group que contendrá el Storage Account del estado. Inicialmente se intentó en eastus pero la suscripción de estudiantes tiene restricciones de región, por lo que se usó southcentralus:

```
az group create -n rg-tfstate-lab8 -l southcentralus
```

```
PS C:\Users\robin> az group create -n rg-tfstate-lab8 -l southcentralus
{
  "id": "/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/rg-tfstate-lab8",
  "location": "southcentralus",
  "managedBy": null,
  "name": "rg-tfstate-lab8",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
```

4b. Storage Account

Antes de crear el Storage Account se verificó que el proveedor Microsoft.Storage estuviera registrado en la suscripción:

```
PS C:\Users\robin> az provider show --namespace Microsoft.Storage --query registrationState -o tsv
NotRegistered
PS C:\Users\robin> az provider register --namespace Microsoft.Storage
Registering is still on-going. You can monitor using 'az provider show -n Microsoft.Storage'
PS C:\Users\robin> az storage account create -g rg-tfstate-lab8 -n sttfstate12345 -l eastus --sku Standard_LRS --encryption-services blob
```

Se creó el Storage Account con SKU Standard_LRS (replicación local) en southcentralus:

```
PS C:\Users\robin> az storage account create -g rg-tfstate-lab8 -n sttfstatelab8robin -l southcentralus --sku Standard_LRS --encryption-services blob
The --min-tls-version argument values TLS1_0 and TLS1_1 have been retired on 2026/02/03 and will be removed on 2026/03/03.
{
  "accessTier": "Hot",
  "accountMigrationInProgress": null,
  "allowBlobPublicAccess": false,
  "allowCrossTenantReplication": false,
  "allowSharedKeyAccess": null,
  "allowedCopyScope": null,
  "azureFilesIdentityBasedAuthentication": null,
  "blobRestoreStatus": null,
  "creationTime": "2026-03-21T06:38:04.840872+00:00",
  "customDomain": null,
  "defaultToOAuthAuthentication": null,
  "dnsEndpointType": null,
  "dualStackEndpointPreference": null,
  "enableExtendedGroups": null,
  "enableHttpsTrafficOnly": true,
  "enableNfsV3": null,
  "encryption": {
    "encryptionIdentity": null,
    "keySource": "Microsoft.Storage",
    "keyVaultProperties": null,
    "requireInfrastructureEncryption": null,
    "services": {
      "blob": {
```

4c. Container

Se creó el container tfstate dentro del Storage Account usando autenticación Azure AD:

```
az storage container create --name tfstate --account-name sttfstatelab8robin --auth-mode login
```

```
PS C:\Users\robin> az storage container create --name tfstate --account-name sttfstatelab8robin --auth-mode login
{
  "created": true
}
PS C:\Users\robin>
```

5. Configuración del Repositorio

5a. Archivo backend.hcl

Se creó el archivo backend.hcl con la configuración del backend remoto. Este archivo no se sube al repositorio por contener datos sensibles de la suscripción:

```
infra > backend.hcl
1 resource_group_name = "rg-tfstate-lab8"
2 storage_account_name = "sttfstatelab8robin"
3 container_name       = "tfstate"
4 key                  = "lab8/terraform.tfstate"
```

5b. Variables de entorno dev.tfvars

Se editó el archivo infra/env/dev.tfvars con la IP pública del equipo de desarrollo para restringir el acceso SSH, y se actualizó la región a southcentralus:

```
PS C:\Users\robin> (Invoke-WebRequest -Uri "https://api.ipify.org" -UseBasicParsing).Content
181.237.38.139
PS C:\Users\robin> |
```

```

...  ← →  Q LAB08_ARSW  ↺  ☰  📱
lb U  variables.tf ...\lb U  main.tf ...\vnet U  outputs.tf ...\vnet U  variables.tf ...\vnet U  .gitignore U
fra > env > dev.tfvars
1  prefix  = "lab8"
2  location = "southcentralus"
3  vm_count = 2
4  admin_username = "student"
5  ssh_public_key = "~/.ssh/id_ed25519.pub"
6  allow_ssh_from_cidr = "181.237.7.90/32"
7  tags = { owner = "robinson", course = "ARSW", env = "dev", expires = "2025-12-31" }
8

```

6. Inicialización y Correcciones de Terraform

6a. Generación de SSH Key

Se generó el par de claves SSH ed25519 necesario para la autenticación en las VMs. La clave privada permanece en el equipo local y la pública se pasa a Terraform:

```
ssh-keygen -t ed25519 -C "robinson.nunez-p@mail.escuelaing.edu.co"
```

```

PS C:\Users\robin> ssh-keygen -t ed25519 -C "robinson.nunez-p@mail.escuelaing.edu.co"
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\robin/.ssh/id_ed25519):
Created directory 'C:\Users\robin/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\robin/.ssh/id_ed25519
Your public key has been saved in C:\Users\robin/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:UVFBhK53ZZ3s12NNDTtMettmwC3QI0Wq50gBVw3M5Ko robinson.nunez-p@mail.escuelaing.edu.co
The key's randomart image is:
+--[ED25519 256]--+
|      ++0==o      |
|      ++ +.=      |
|      . o.o .B.*o  |
|      o *... @o=   |
|      S.o  +.Bo    |
|      o...  .+B    |
|      E ..  .oo    |
|                  |
+-----[SHA256]-----+
PS C:\Users\robin>

```

6b. Corrección de módulos duplicados

Al intentar inicializar Terraform se detectaron variables y outputs duplicados entre los archivos main.tf y los archivos variables.tf/outputs.tf de los módulos. Se corrigieron los siguientes archivos:

- En modules/lb/main.tf: se eliminaron las declaraciones de variables al inicio del archivo

```

Archivo  Editar  Ver

resource "azurerm_public_ip" "pip" {
  name                = "${var.prefix}-lb-pip"
  location            = var.location
  resource_group_name = var.resource_group_name
  allocation_method   = "Static"
  sku                 = "Standard"
  tags               = var.tags
}

resource "azurerm_lb" "lb" {
  name                = "${var.prefix}-lb"
  location            = var.location
  resource_group_name = var.resource_group_name
  sku                 = "Standard"

  frontend_configuration {
    name                = "Public"
    public_ip_address_id = azurerm_public_ip.pip.id
  }
  tags = var.tags
}

resource "azurerm_lb_backend_address_pool" "bepool" {
  name                = "${var.prefix}-bepool"
  loadbalancer_id     = azurerm_lb.id
}

# Asociar NICs al backend pool
resource "azurerm_network_interface_backend_address_pool_association" "assoc" {
  count               = length(var.backend_nic_ids)
  network_interface_id = var.backend_nic_ids[count.index]
  ip_configuration_name = "ipconfig1"
  backend_address_pool_id = azurerm_lb_backend_address_pool.bepool.id
}

```

- En modules/vnet/main.tf: se eliminó el output duplicado al final del archivo

```
resource "azurerm_virtual_network" "vnet" {
  name                = "${var.prefix}-vnet"
  address_space       = ["10.10.0.0/16"]
  location            = var.location
  resource_group_name = var.resource_group_name
  tags               = var.tags
}

resource "azurerm_subnet" "web" {
  name                = "subnet-web"
  resource_group_name = var.resource_group_name
  virtual_network_name = azurerm_virtual_network.vnet.name
  address_prefixes     = ["10.10.1.0/24"]
}

resource "azurerm_subnet" "mgmt" {
  name                = "subnet-mgmt"
  resource_group_name = var.resource_group_name
  virtual_network_name = azurerm_virtual_network.vnet.name
  address_prefixes     = ["10.10.2.0/24"]
}

output "subnet_web_id" {
  value = azurerm_subnet.web.id
}
```

- Se eliminaron los archivos outputs.tf duplicados de los módulos lb y vnet

```
variable "resource_group_name" { type = string }
variable "location" { type = string }
variable "prefix" { type = string }
variable "admin_username" { type = string }
variable "ssh_public_key" { type = string }
variable "subnet_id" { type = string }
variable "vm_count" { type = number }
variable "cloud_init" { type = string }
variable "tags" { type = map(string) }

resource "azurerm_network_interface" "nic" {
  count                = var.vm_count
  name                = "${var.prefix}-nic-${count.index}"
  location            = var.location
  resource_group_name = var.resource_group_name

  ip_configuration {
    name                = "ipconfig1"
    subnet_id          = var.subnet_id
    private_ip_address_allocation = "Dynamic"
  }
  tags = var.tags
}

resource "azurerm_linux_virtual_machine" "vm" {
  count                = var.vm_count
  name                = "${var.prefix}-vm-${count.index}"
  location            = var.location
  resource_group_name = var.resource_group_name
  size                = "Standard_B1s"

  admin_username = var.admin_username
  network_interface_ids = [azurerm_network_interface.nic[count.index].id]

  admin_ssh_key {
    username   = var.admin_username
    public_key = var.ssh_public_key
  }
}
```

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> Remove-Item ..\modules\lb\outputs.tf
PS C:\Users\robin\downloads\LAB08_ARSW\infra> Remove-Item ..\modules\vnet\outputs.tf
```

6c. terraform init

Una vez corregidos los módulos se ejecutó la inicialización con el backend remoto:

```
terraform init -backend-config backend.hcl
```

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform init -backend-config backend.hcl
Initializing the backend...
Initializing modules...
Initializing provider plugins...
- Finding hashicorp/azurerm versions matching ">= 4.0"...
- Installing hashicorp/azurerm v4.65.0...
- Installed hashicorp/azurerm v4.65.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Users\robin\downloads\LAB08_ARSW\infra> |
```

6d. Recreación de variables perdidas

Durante las correcciones se perdió el archivo variables.tf del módulo compute. Se recreó con todas las variables necesarias:

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform init -backend-config backend.hcl
Initializing the backend...
Initializing modules...
Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 4.0"...
- Installing hashicorp/azurerm v4.65.0...
- Installed hashicorp/azurerm v4.65.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform fmt -recursive
env\dev.tfvars
```

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> @"
>> variable "resource_group_name" { type = string }
>> variable "location"            { type = string }
>> variable "prefix"              { type = string }
>> variable "admin_username"      { type = string }
>> variable "ssh_public_key"      { type = string }
>> variable "subnet_id"          { type = string }
>> variable "vm_count"            { type = number }
>> variable "cloud_init"          { type = string }
>> variable "tags"                { type = map(string) }
>> "@ | Out-File -FilePath ..\modules\compute\variables.tf -Encoding utf8
PS C:\Users\robin\downloads\LAB08_ARSW\infra> cat ..\modules\compute\variables.tf
variable "resource_group_name" { type = string }
variable "location"           { type = string }
variable "prefix"             { type = string }
variable "admin_username"     { type = string }
variable "ssh_public_key"     { type = string }
variable "subnet_id"          { type = string }
variable "vm_count"           { type = number }
variable "cloud_init"         { type = string }
variable "tags"               { type = map(string) }
```

Se recreó también el outputs.tf del módulo vnet:

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> @"
>> output "vnet_name"           { value = azurerm_virtual_network.vnet.name }
>> output "subnet_web_id"       { value = azurerm_subnet.web.id }
>> "@ | Out-File -FilePath ..\modules\vnet\outputs.tf -Encoding utf8
```

6e. Validación

Se verificó que la configuración de Terraform es válida:

```
terraform validate
```

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform validate
Success! The configuration is valid.
```

7. Plan y Despliegue de Infraestructura

7a. Terraform Plan

Se generó el plan de ejecución para revisar los 18 recursos que se crearían antes de aplicar:

```
terraform plan -var-file env/dev.tfvars -out plan.tfplan
```

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform plan -var-file env/dev.tfvars -out plan.tfplan
Acquiring state lock. This may take a few moments...

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
+ create

Terraform will perform the following actions:

# azure_rm_resource_group.rg will be created
+ resource "azure_rm_resource_group" "rg" {
+   id           = (known after apply)
+   location     = "southcentralus"
+   name         = "lab8-rg"
+   tags        = {
+     "course" = "ARSW"
+     "env"    = "dev"
+     "expires" = "2025-12-31"
+     "owner"  = "robinson"
+   }
}

# module.compute.azure_rm_linux_virtual_machine.vms[0] will be created
+ resource "azure_rm_linux_virtual_machine" "vm" {
+   admin_username      = "student"
+   allow_extension_operations = (known after apply)
```

```
+   guid                = (known after apply)
+   id                  = (known after apply)
+   location            = "southcentralus"
+   name                 = "lab8-vnet"
+   private_endpoint_vnet_policies = "Disabled"
+   resource_group_name = "lab8-rg"
+   subnet               = (known after apply)
+   tags                = {
+     "course" = "ARSW"
+     "env"    = "dev"
+     "expires" = "2025-12-31"
+     "owner"  = "robinson"
+   }
}

Plan: 18 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ lb_public_ip      = (known after apply)
+ resource_group_name = "lab8-rg"
+ vm_names          = [
+   "lab8-vm-0",
+   "lab8-vm-1",
+ ]

Saved the plan to: plan.tfplan

To perform exactly these actions, run the following command to apply:
  terraform apply "plan.tfplan"
Releasing state lock. This may take a few moments...
PS C:\Users\robin\downloads\LAB08_ARSW\infra> |
```

7b. Problema de capacidad de VM

Al aplicar el plan, Azure rechazó la creación de las VMs con tamaño Standard_B1s por restricciones de capacidad en la región southcentralus. Se investigaron los tamaños disponibles y se identificó que solo los tamaños ARM64 serie v2 estaban disponibles sin restricciones.

Se modificó el tamaño de VM a Standard_B2pls_v2 y se actualizó la imagen de Ubuntu de 22_04-lts-gen2 (x64) a 22_04-lts-arm64 para compatibilidad:


```

PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform apply "plan.tfplan"
Acquiring state lock. This may take a few moments...
azurerm_resource_group.rg: Creating...
azurerm_resource_group.rg: Still creating... [00m10s elapsed]
azurerm_resource_group.rg: Still creating... [00m20s elapsed]
azurerm_resource_group.rg: Creation complete after 27s [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg]
module.vnet.azure_rm_virtual_network.vnet: Creating...
module.lb.azure_rm_public_ip.pip: Creating...
module.lb.azure_rm_network_security_group.nsg: Creating...
module.lb.azure_rm_public_ip.pip: Creation complete after 6s [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/publicIPAddresses/lab8-lb-pip]
module.lb.azure_rm_lb.lb: Creating...
module.lb.azure_rm_network_security_group.nsg: Creation complete after 6s [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/networkSecurityGroups/lab8-web-nsg]
module.vnet.azure_rm_virtual_network.vnet: Creation complete after 9s [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/virtualNetworks/lab8-vnet]
module.vnet.azure_rm_subnet.web: Creating...
module.vnet.azure_rm_subnet.mgmt: Creating...
module.vnet.azure_rm_subnet.mgmt: Creation complete after 6s [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/virtualNetworks/lab8-vnet/subnets/subnet-mgmt]
module.lb.azure_rm_lb.lb: Still creating... [00m10s elapsed]
module.vnet.azure_rm_subnet.web: Still creating... [00m10s elapsed]
module.lb.azure_rm_lb.lb: Creation complete after 13s [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/loadBalancers/lab8-lb]

```

7c. Apply exitoso

Con los cambios aplicados, el despliegue completó exitosamente los 18 recursos en aproximadamente 2 minutos:

```

resource "azurerm_network_interface" "nic" {
  count                = var.vm_count
  name                 = "${var.prefix}-nic-${count.index}"
  location              = var.location
  resource_group_name = var.resource_group_name

  ip_configuration {
    name                 = "ipconfig1"
    subnet_id            = var.subnet_id
    private_ip_address_allocation = "Dynamic"
  }
  tags = var.tags
}

resource "azurerm_linux_virtual_machine" "vm" {
  count                = var.vm_count
  name                 = "${var.prefix}-vm-${count.index}"
  location              = var.location
  resource_group_name = var.resource_group_name
  size                 = "Standard_B2pls_v2"

  admin_username = var.admin_username
  network_interface_ids = [azurerm_network_interface.nic[count.index].id]

  admin_ssh_key {
    username = var.admin_username
    public_key = var.ssh_public_key
  }

  source_image_reference {
    publisher = "Canonical"
    offer     = "0001-com-ubuntu-server-jammy"
    sku       = "22 04-lts-arm64"
    version   = "latest"
  }

  custom_data = base64encode(var.cloud_init)
}

```

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform apply "plan.tfplan"
Acquiring state lock. This may take a few moments...
module.compute.azure_terraform_virtual_machine.vm[0]: Creating...
module.compute.azure_terraform_virtual_machine.vm[1]: Creating...
module.compute.azure_terraform_virtual_machine.vm[0]: Still creating... [00m10s elapsed]
module.compute.azure_terraform_virtual_machine.vm[1]: Still creating... [00m10s elapsed]
module.compute.azure_terraform_virtual_machine.vm[1]: Creation complete after 19s [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Compute/virtualMachines/lab8-vm-1]
module.compute.azure_terraform_virtual_machine.vm[0]: Still creating... [00m20s elapsed]
module.compute.azure_terraform_virtual_machine.vm[0]: Still creating... [00m30s elapsed]
module.compute.azure_terraform_virtual_machine.vm[0]: Still creating... [00m40s elapsed]
module.compute.azure_terraform_virtual_machine.vm[0]: Still creating... [00m50s elapsed]
module.compute.azure_terraform_virtual_machine.vm[0]: Creation complete after 51s [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Compute/virtualMachines/lab8-vm-0]
Releasing state lock. This may take a few moments...

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:
lb_public_ip = "20.114.114.201"
resource_group_name = "lab8-rg"
vm_names = [
  "lab8-vm-0",
  "lab8-vm-1",
]
PS C:\Users\robin\downloads\LAB08_ARSW\infra> |
```

Output/Valor

lb_public_ip	20.114.114.201
resource_group_name	lab8-rg
vm_names	lab8-vm-0, lab8-vm-1

8. Verificación del Load Balancer

Se verificó que el Load Balancer distribuye el tráfico correctamente entre las dos VMs ejecutando curl contra la IP pública. Cada VM responde con su hostname, demostrando que nginx está funcionando y el balanceo opera correctamente:

Respuesta de lab8-vm-1

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> curl http://20.114.114.201

Advertencia de seguridad: riesgo de ejecución de script
Invoke-WebRequest analiza el contenido de la página web. El código de script de la página web se puede ejecutar cuando se analiza
página.
  ACCIÓN RECOMENDADA:
    Usa el modificador -UseBasicParsing para evitar la ejecución de código de script.

¿Quieres continuar?

[S] Sí [O] Sí a todo [N] No [T] No a todo [U] Suspender [?] Ayuda (el valor predeterminado es "N"): S

StatusCode      : 200
StatusDescription : OK
Content          : Hola desde lab8-vm-1
RawContent       : HTTP/1.1 200 OK
                  Connection: keep-alive
                  Accept-Ranges: bytes
                  Content-Length: 21
                  Content-Type: text/html
                  Date: Sat, 21 Mar 2026 19:14:04 GMT
                  ETag: "69beeca9-15"
                  Last-Modified: Sat, 21 Mar 2026 19...
Forms            : {}
Headers          : {[Connection, keep-alive], [Accept-Ranges, bytes], [Content-Length, 21], [Content-Type, text/html]...}
Images           : {}
InputFields      : {}
Links            : {}
ParsedHtml       : mshtml.HTMLDocumentClass
RawContentLength : 21
```

Respuesta de lab8-vm-0

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> 1..4 | ForEach-Object { (Invoke-WebRequest -Uri "http://20.114.114.201" -UseBasicParsing).Content }
Hola desde lab8-vm-0

Hola desde lab8-vm-0

Hola desde lab8-vm-0

Hola desde lab8-vm-0
```

Ambas máquinas respondiendo

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> Write-Host "Load Balancer IP: 20.114.114.201"
Load Balancer IP: 20.114.114.201
PS C:\Users\robin\downloads\LAB08_ARSW\infra> Write-Host "--- Respuesta VM ---"
--- Respuesta VM ---
PS C:\Users\robin\downloads\LAB08_ARSW\infra> (Invoke-WebRequest -Uri "http://20.114.114.201" -UseBasicParsing).Content
Hola desde lab8-vm-0

PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform output
lb_public_ip = "20.114.114.201"
resource_group_name = "lab8-rg"
vm_names = [
  "lab8-vm-0",
  "lab8-vm-1",
]
PS C:\Users\robin\downloads\LAB08_ARSW\infra> |
```

Evidencia de balanceo

El Load Balancer distribuye las peticiones HTTP entre lab8-vm-0 y lab8-vm-1. Cada VM responde "Hola desde lab8-vm-X" confirmando que nginx está activo y el health probe TCP/80 pasa correctamente.

9. GitHub Actions con OIDC

Se configuró un pipeline de CI/CD usando GitHub Actions con autenticación OIDC (OpenID Connect) hacia Azure. Esta aproximación elimina la necesidad de secretos de larga duración (client secrets) al federar la identidad del repositorio directamente con Azure AD.

9a. Creación de la App en Azure AD

Se creó el App Registration que representará al workflow de GitHub Actions ante Azure AD:

```
az ad app create --display-name "github-terraform-lab8"
```

appId obtenido: 7e0195b6-9293-401d-a5bc-34f49183f5ae

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> az ad app create --display-name "github-terraform-lab8"
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#applications/$entity",
  "addIns": [],
  "api": {
    "acceptMappedClaims": null,
    "knownClientApplications": [],
    "oauth2PermissionScopes": [],
    "preAuthorizedApplications": [],
    "requestedAccessTokenVersion": null
  },
  "appId": "7e0195b6-9293-401d-a5bc-34f49183f5ae",
  "appNotes": [],
  "applicationTemplateId": null,
  "certification": null,
  "createdByAppId": "04b07795-8ddb-461a-bbee-02f9e1bf7b46",
  "createdDateTime": "2026-03-21T19:20:28.1685853Z",
  "defaultRedirectUri": null,
  "deletedDateTime": null,
  "description": null,
  "disabledByMicrosoftStatus": null,
  "displayName": "github-terraform-lab8",
  "groupMembershipClaims": null,
  "id": "1c477b7f-ec6e-4fd1-b289-7a2bb6f515eb",
  "identifierUri": [],
  "info": {
    "logoUrl": null,
    "marketingUrl": null,
    "privacyStatementUrl": null,
    "supportUrl": null,
    "termsOfServiceUrl": null
  },
}
```

Se creó el Service Principal asociado a la app:

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> az ad sp create --id 7e0195b6-9293-401d-a5bc-34f49183f5ae
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#servicePrincipals/$entity",
  "accountEnabled": true,
  "addIns": [],
  "alternativeNames": [],
  "appDescription": null,
  "appDisplayName": "github-terraform-lab8",
  "appId": "7e0195b6-9293-401d-a5bc-34f49183f5ae",
  "appOwnerOrganizationId": "50640584-2a40-4216-a84b-9b3ee0f3f6cf",
  "appRoleAssignmentRequired": false,
  "appRoles": [],
  "applicationTemplateId": null,
  "createdByAppId": "04b07795-8ddb-461a-bbee-02f9e1bf7b46",
  "createdDateTime": "2026-03-21T19:22:57Z",
  "deletedDateTime": null,
  "description": null,
  "disabledByMicrosoftStatus": null,
  "displayName": "github-terraform-lab8",
  "homepage": null,
  "id": "89a23f52-fc8b-4320-966f-098121d16151",
  "info": {
    "logoUrl": null,
    "marketingUrl": null,
    "privacyStatementUrl": null,
    "supportUrl": null,
    "termsOfServiceUrl": null
  }
}
```

9b. Asignación del rol Contributor

Se asignó el rol Contributor al Service Principal sobre la suscripción completa, permitiéndole crear y gestionar los recursos del lab:

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> az role assignment create `
>> --assignee 7e0195b6-9293-401d-a5bc-34f49183f5ae `
>> --role Contributor `
>> --scope /subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a
{
  "condition": null,
  "conditionVersion": null,
  "createdBy": null,
  "createdOn": "2026-03-21T19:24:51.504830+00:00",
  "delegatedManagedIdentityResourceId": null,
  "description": null,
  "id": "/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/providers/Microsoft.Authorization/roleAssignments/80dfcff5-2e36-4926-89c5-af409cdabdd0",
  "name": "80dfcff5-2e36-4926-89c5-af409cdabdd0",
  "principalId": "89a23f52-fc8b-4320-966f-098121d16151",
  "principalType": "ServicePrincipal",
  "roleDefinitionId": "/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/providers/Microsoft.Authorization/roleDefinitions/b24988ac-6180-42a0-ab88-20f7382dd24c",
  "scope": "/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a",
  "type": "Microsoft.Authorization/roleAssignments",
  "updatedBy": "db6ff7fb-bbc9-4233-bd7f-80137cf497d8",
  "updatedOn": "2026-03-21T19:24:51.813819+00:00"
}
PS C:\Users\robin\downloads\LAB08_ARSW\infra>
```

9c. Federación OIDC

Se configuró la federación entre el repositorio GitHub y Azure AD, especificando que solo el branch main puede obtener tokens:

Repositorio: https://github.com/Robinson677/LAB08_ARSW

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> az ad app federated-credential create `
>> --id 7e0195b6-9293-401d-a5bc-34f49183f5ae `
>> --parameters '{"name":"github-oidc","issuer":"https://token.actions.githubusercontent.com","subject":"repo:Robinson677/LAB08_ARSW:ref:refs/heads/main","audiences":["api://AzureADTokenExchange"]}'
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#applications('1c477b7f-ec6e-4fd1-b289-7a2bb6f515eb')/federatedIdentities/$entity",
  "audiences": [
    "api://AzureADTokenExchange"
  ],
  "description": null,
  "id": "9d74ea60-c2cf-46cd-89a4-2c6734ca01e5",
  "issuer": "https://token.actions.githubusercontent.com",
  "name": "github-oidc",
  "subject": "repo:Robinson677/LAB08_ARSW:ref:refs/heads/main"
}
PS C:\Users\robin\downloads\LAB08_ARSW\infra>
```

9d. Workflow de GitHub Actions

Se creó el archivo `.github/workflows/terraform.yml` con dos jobs: plan automático en cada push y apply manual via `workflow_dispatch`. El workflow ejecuta `fmt`, `validate` y `plan` automáticamente:

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> cd C:\Users\robin\downloads\LAB08_ARSW
PS C:\Users\robin\downloads\LAB08_ARSW> New-Item -ItemType Directory -Force -Path .github/workflow

Directorio: C:\Users\robin\downloads\LAB08_ARSW\.github

Mode                LastWriteTime         Length Name
----                -
d-----          21/03/2026  2:32 p. m.             workflow

PS C:\Users\robin\downloads\LAB08_ARSW> notepad .github/workflows/terraform.yml
```

```
name: Terraform CI/CD

on:
  push:
    branches: [main]
  pull_request:
    branches: [main]
  workflow_dispatch:

permissions:
  id-token: write
  contents: read

env:
  ARM_CLIENT_ID: ${ secrets.AZURE_CLIENT_ID }
  ARM_TENANT_ID: ${ secrets.AZURE_TENANT_ID }
  ARM_SUBSCRIPTION_ID: ${ secrets.AZURE_SUBSCRIPTION_ID }
  ARM_USE_OIDC: true

jobs:
  plan:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

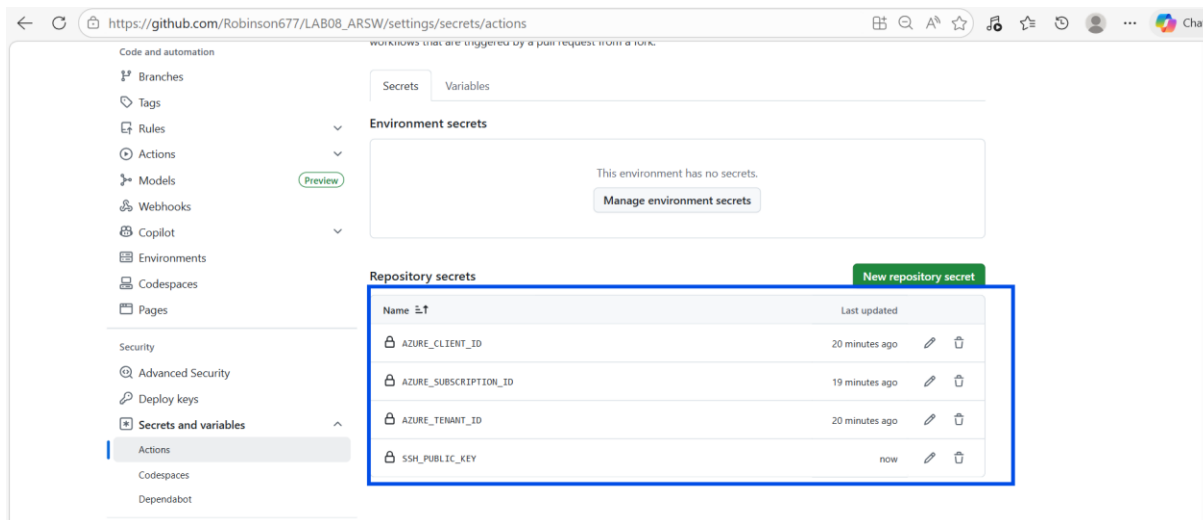
      - name: Azure login
        uses: azure/login@v2
        with:
          client-id: ${ secrets.AZURE_CLIENT_ID }
          tenant-id: ${ secrets.AZURE_TENANT_ID }
          subscription-id: ${ secrets.AZURE_SUBSCRIPTION_ID }

      - name: Setup Terraform
        uses: hashicorp/setup-terraform@v3
        with:
          terraform_version: "1.6.0"

      - name: Terraform Init
```

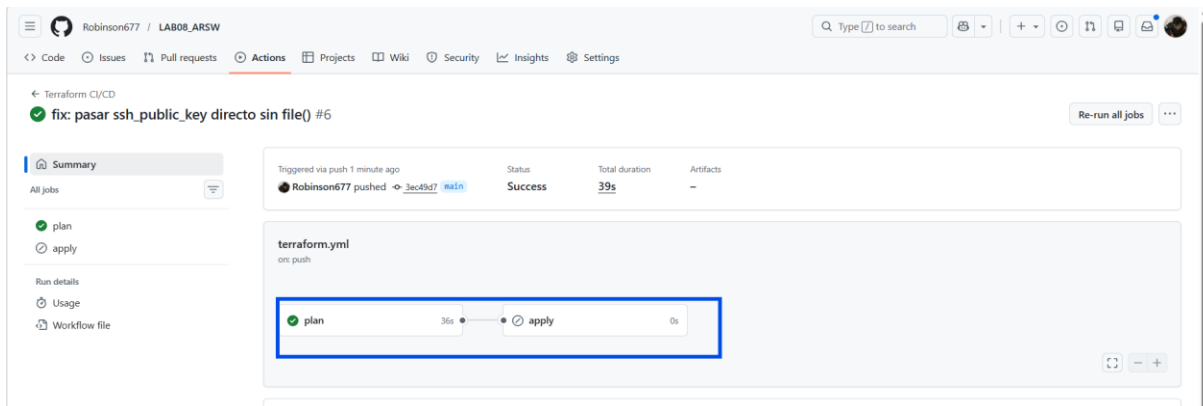
9e. Secretos en GitHub

Se configuraron los tres secretos necesarios más la clave SSH pública en el repositorio:



9f. Ejecución del Pipeline

Luego de resolver múltiples ajustes como eliminación de outputs duplicados, inline del backend.hcl, paso de SSH key como secreto, y corrección del file() en main.tf, el pipeline ejecutó exitosamente con el job plan en verde:



Pipeline exitoso

El job plan ejecuta fmt, validate y plan automáticamente. El job apply se omite en push solo se activa manualmente con workflow_dispatch, comportamiento correcto según el diseño del pipeline.